

COMPUTER SCIENCE Journey on a Page

Curriculum Intent

The intent of the Computer Science curriculum is to enable all students to develop an elevated level of IT skills and knowledge to ensure that they are confident and assured within their own digital literacy for their step-in education, training or employment.

The core of the subject is the understanding of how technology works, can be developed and utilised.

Students will be able to contextualise theory and knowledge learnt in the lessons. They should know the how and why around computer system concepts. They will also develop their skills as programmers, becoming logical, problem solving and resilient young people in the process. We aim for students to become resilient problem solvers and relish a challenge.

MEA Computer Science learn to:

- ✓ Understand and the fundamental principles and concepts of Computer Science
- ✓ Analyse problems in computational terms.
- ✓ Understand the components in digital systems.
- ✓ Use mathematics where possible to help find solutions to problems.
- ✓ Think creatively, innovatively, analytically, logically, and critically.
- ✓ Look for more than one solution to any problem.
- ✓ Understand the impacts of digital technology on an individual, national, and global level. Question what and why, not to rely on conceptualisation

Threshold Concepts | Students will *know* and *understand* that...

	Computer Systems	Networks	Impacts of IT	Algorithms and Logic	Programming	Software Development
Year 10	1.1 Systems architecture 1.2 Memory and storage 1.5 Systems software	1.3 Computer networks, connections, and protocols 1.4 Network security	1.6 Ethical, legal, cultural, and environmental impacts of digital technology	2.1 Algorithms 2.4 Boolean logic	2.2 Programming fundamentals	2.3 Producing robust programs. 2.5 Programming languages and Integrated Development Environments
Year 11	1.1 Systems architecture 1.2 Memory and storage 1.5 Systems software	1.3 Computer networks, connections, and protocols 1.4 Network security	1.6 Ethical, legal, cultural, and environmental impacts of digital technology	2.1 Algorithms 2.4 Boolean logic	2.2 Programming fundamentals	2.3 Producing robust programs. 2.5 Programming languages and Integrated Development Environments

COMPUTER SCIENCE Journey on a Page

Note:

The threshold concepts above are taught in year 10. This means the whole specification is covered in year 10. Every piece of procedural knowledge below is gained in year 10, then strengthened in year 11. During year 11, we recap knowledge. We introduce more exam questions which are of greater difficulty.

Procedural Knowledge

Students will learn and practise *how to...*

Computer Systems

1.1 Systems Architecture

- Annotate the components in the CPU.
- Narrate the stages of the FDE.
- Suggest a range of embedded systems.

1.2 Memory and Storage

- Describe the differences between RAM and ROM
- Order units of data from smallest to largest
- Convert data types.
- Add binary numbers.
- Convert between number systems.
- Calculate required storage capacity for a given set of files.
- Carry out a binary shift (both left and right)
- Calculate bit depth and the sample rate of an audio file.
- Recommend and justify storage devices for a specific context.
- Recommend and justify a type of compression devices for a specific context.

1.3 Computer networks, connections, and protocols

- Annotate a network topology.
- Convery how a DNS works in the form of a flowchart of algorithm.
- Contextualise types of computer network and network components for a given scenario
- Recommend network protocols within a scenario.

1.4 Network Security

Computational thinking, algorithms, and Programming

2.1 Algorithms

- Produce and edit flowcharts.
- Create and use trace tables.
- Complete, write or refine an algorithm using the techniques listed.
- Use nesting and selection.
- Apply an algorithm to a data set.
- Express a solution in reference language or a flowchart.

2.2 Programming fundamentals

- Use operators within programming.
- Program the following:
 - Concatenation
 - Slicing
 - Variables
 - Casting
 - 1D/2D arrays
 - Functions
 - Procedures
 - SQL:
 - SELECT / WHERE / FROM
 - Wildcards
 - Random numbers
 - File operations

COMPUTER SCIENCE Journey on a Page

- Articulate forms of network attack
- Articulate common prevention methods against threats to a network.

1.5 Systems Software

- Recommend system and utility software.
- Choose a full or incremental backup, based on a scenario.
- Annotate how encryption works.

1.6 Ethical, legal, cultural, and environmental impacts of digital technology

- Structure a 6/8-mark exam question
- Write extended answers around the ethical, legal, cultural, environmental and privacy issues which effect computing.

2.3 Producing robust programs.

- Identify syntax/logic errors in code and suggest fixes.
- Test code for logic and syntax errors
- Identify suitable test data for a scenario.
- Create a test plan.
- Test data using the three types of test values:
 - Standard
 - Erroneous
 - Boundary

2.4 Boolean logic

- Draw a truth table.
- Draw a logic diagram.

2.5 Programming languages and Integrated Development Environments

- Recommend high- or low-level languages.
- Know how to use one IDE or more

COMPUTER SCIENCE Big Picture Year 10

Autumn 1	Autumn 2
Threshold Concepts and Procedural Knowledge J277/01: Computer Systems: 1.1 Systems architecture 1.1.1 Architecture of the CPU 1.1.2 CPU performance 1.1.3 Embedded systems 1.2 Memory and storage 1.2.1 Primary storage (Memory) 1.2.2 Secondary storage 1.2.3 Units 1.2.4 Data storage 1.2.5 Compression Threshold Concepts: - Computer Systems	Threshold Concepts and Procedural Knowledge J277/01: Computer Systems: 1.3 Computer networks, connections and protocols 1.3.1 Networks and topologies 1.3.2 Wired and wireless networks, protocols and layers 1.4 Network security 1.4.1 Threats to computer systems and networks 1.4.2 Identifying and preventing vulnerabilities Threshold Concepts: - Networks
Assessment Mini test in the last week of half term on AO 1.1 and 1.2.	Assessment Mini test in the last week of half term on AO 1.3 and 1.4.
Spring 1	Spring 2
Threshold Concepts and Procedural Knowledge J277/01: Computer Systems: 1.5 Systems software 1.5.1 Operating systems 1.5.2 Utility software 1.6 Ethical, legal, cultural and environmental impacts of digital technology Threshold Concepts: - Computer Systems - Impact of IT	Threshold Concepts and Procedural Knowledge J277/02: Computational thinking, algorithms and Programming: 2.1 Algorithms 2.1.1 Computational thinking 2.1.2 Designing, creating and refining algorithms 2.1.3 Searching and sorting algorithms 2.2 Programming fundamentals 2.2.1 Programming fundamentals Threshold Concepts: - Software Development - Programming - Algorithms

COMPUTER SCIENCE Big Picture Year 10

Assessment Mini test in the first week of half term on AO 1.1, 1.2, 1.3 and 1.4 Progress Exam 1	Assessment Paper 1 mini test. (60 marks – to be done in lesson).
Summer 1	Summer 2
Threshold Concepts and Procedural Knowledge J277/02: Computational thinking, algorithms and Programming: 2.2.1 Programming fundamentals 2.2.2 Data types 2.2.3 Additional programming techniques 2.3 Producing robust programs 2.3.1 Defensive design 2.3.2 Testing 2.4 Boolean logic 2.4.1 Boolean Logic Threshold Concepts: • Software Development • Programming • Algorithms	Threshold Concepts and Procedural Knowledge J277/02: Computational thinking, algorithms and Programming: 2.5 Programming languages and Integrated Development Environments 2.5.1 Languages 2.5.2 The Integrated Development Environment Threshold Concepts: • Software Development
Assessment Mini test in the first week of half term on AO 2.1. Mini test in the last week of half term on AO 2.2.	Assessment Progress Exam 2